

MUTWALLI ALSHEIKH

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AI/ML Engineer Intern | Computer Science Student, Graduating Dec 2027 | U.S. Citizen — No Sponsorship Required

EDUCATION

United Arab Emirates University

Sep 2023 – Expected Dec 2027

B.S. in Computer Science — GPA: 3.6/4.0 • Dean's List (x3)

Relevant Coursework: Artificial Intelligence, Machine Learning, Data Structures & Algorithms, Linear Algebra, Probability & Statistics

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript

Machine Learning: PyTorch, scikit-learn, LightGBM, Hugging Face Transformers, sentence-transformers, NumPy, pandas, SHAP, NLP

LLM Systems: RAG pipelines, embeddings, vector search (Qdrant), tool calling, agent evaluation, Llama 3.1, Ollama

Backend & Web: FastAPI, React, Next.js, PostgreSQL, SQLAlchemy, Alembic, Redis, Docker, Docker Compose, GitHub Actions, Render, Vercel, Git

EXPERIENCE

Undergraduate Researcher — SURE Plus Program, UAEU

May 2024 – Oct 2024

- Designed the statistical evaluation framework for a faculty-supervised mobile crowdsensing study, analyzing 50K+ sensor observations across real-world mobility scenarios.
- Engineered automated Python evaluation pipelines for crowdsensing algorithms, cutting analysis time ~30% and enabling rapid iteration across experimental conditions.

Mathematics Tutor — United Arab Emirates University

Jan 2024 – Present

- Tutor undergraduates across Calculus I-III, Linear Algebra, and Probability & Statistics; build targeted practice sets around recurring misconceptions.

PROJECTS

AutoPilot — Autonomous LLM Agent Platform | Python, FastAPI, Next.js, PostgreSQL, Redis, Docker | Live demo

- Built a ReAct-style agent over 6 registered tools with structured task planning, human approval gates on privileged actions, concurrency-limited parallel execution, and retrieval-backed memory over embedded execution history.
- Defined four failure classes mapped to retry, replan, or halt, persisting each step as a structured trace so reconnecting WebSocket clients replay complete execution history from the database.
- Scored 22 of 26 tasks (85%) on a self-designed benchmark across 5 categories at \$0.84 total inference cost; hardened with 564 tests across 35 modules under dual-database CI (SQLite + PostgreSQL).

RAG Filing Analyst | Python, FastAPI, React, Qdrant, Llama 3.1

- Shipped a RAG pipeline over SEC 10-K filings using all-MiniLM-L6-v2 embeddings in Qdrant and fully local Llama 3.1-8B inference via Ollama, returning source-grounded answers with citations and no external API calls; containerized as a 4-service Docker Compose stack with GitHub Actions CI and 83 tests.
- Raised Hit@1 from 0.76 to 0.92 and nDCG@10 from 0.83 to 0.96 by adding BM25 sparse retrieval fused server-side with reciprocal rank fusion, measured on a 50-query hand-verified gold set and confirmed by paired bootstrap (+0.131 nDCG, p=0.001).
- Measured cross-encoder reranking at no significant gain (p=0.69) for 15x the retrieval latency and left it disabled; eliminated unstable RRF tie ordering that returned reshuffled citations for identical queries.

Market-Regime Forecasting & Backtesting Pipeline | Python, LightGBM, scikit-learn, SHAP, Streamlit

- Built an expanding-window walk-forward pipeline over 5 folds and 1,260 out-of-sample trading days, with separate LightGBM classification and regression models, 4 naive baselines, and 3 regime features (volatility, trend, drawdown).
- Validated it against four baselines and found no predictive edge: out-of-sample ROC-AUC was 0.504, and a 20-day rolling-mean baseline led on annualized return/volatility (0.91) over buy-and-hold (0.54) and the best LightGBM model (0.45). Fixed a bug understating transaction costs 62x and a target leak at fold boundaries, locking both down with 96 assertions validated by mutation testing.

Adaptive Duelist AI | Python, scikit-learn, Pygame, FastAPI, SQLite

- Built a real-time game opponent predicting a player's next action from an event-triggered Markov-chain and Random Forest ensemble across 7 tactical modes and 5 archetypes, unified behind one simulation step shared by the engine, replay verifier, evaluator, and test harness.
- Ran a 3-tier ablation over 400 identically seeded matches per tier, cutting mean time to defeat the opponent from 3,559 to 980 ticks; instrumentation showed 100% of top-tier actions came from a hand-written reaction layer, while top-1 prediction accuracy (24.4%) trailed the majority-class baseline (32.2%).
- Fixed cross-process nondeterminism from unordered enum sets under Python hash randomization, restoring all 14 replay verification checks; mutation testing then exposed 6 uncovered failure paths, closed with 24 additional tests (1,063 total).

AWARDS & LEADERSHIP

Awards: UAEU Hackathon 3rd (2025) • UAEU Chess Tournament 1st (2024) • UAE Inter-University Soccer League — 1st (2024).

Leadership & Activities: Hackathon Community Leader • Chess Club Organizer • Varsity Soccer & Basketball